



Motorola's Freecharge. Charge on the go!

research into hydrogen fuel cells. Most of this research will look into larger applications such as replacing petrol engines in cars. Private companies are falling over themselves to develop fuel cells for portable devices as these would command a high premium in the relatively affluent mobile phone and laptop markets.

What else can be done?

One age-old solution to the failing battery is the solar panel idea. For years, calculators

have had dual power sources—button batteries and a solar panel. The concept actually works very well for such low power devices, but today's high tech cell phones and laptops require a really large surface filled with solar cells to generate enough power to run them, defeating the idea of miniaturisation. A new type of solar cell tackles that problem. It is inexpensive and flexible, so that you can integrate it into your clothes which occupy a huge surface area. Scientists from the University of California at Berkeley have made a hybrid nanorod-polymer solar cell which is about one-thousandth as wide as a human hair. The cells could be spin-cast onto almost any material, including fabric, as they are plastic-based, but deliver only 2 per cent efficiency—as compared to 30 per cent by today's cutting edge solar cells. The nanorod-polymer cells can't power your devices but they can provide an auxiliary power source to charge batteries slowly.

Another way to make batteries better is to simply make them charge faster. American battery maker Rayovac has developed a method to fully charge NiMH batteries in just 15 minutes—and getting to 90 per

cent of full charge in just 10 minutes. This is achieved by adding battery monitoring, called 'in-cell charge control,' which monitors the state of the chemical reaction during charging. The batteries will also give better performance than standard ones. They will have a capacity of 2,000 mAh compared to the normal 1,600 mAh for AA cells. While fast charging is definitely a plus, it's totally useless if you simply have no place at all to plug in.

Motorola addresses just that with its hand-cranked charger for cell phones. Known as FreeCharge, it

retails for \$50 (Rs 2,350) and is based on technology from a company called FreePlay. A handle drives a generator, which charges its own NiMH battery, which in turn charges your phone. The internal battery takes 35 minutes of winding to charge fully. Thankfully, you can wind for only as long as you want, not the full 35 minutes. The company reckons you can get 4 to 6 minutes of talk time out of a 45 second work out.

A bizarre idea thought up by an engineer at Compaq uses human power, but no extra effort. The idea is to use your keystrokes to provide recharging power for devices such as laptops. This could either reduce the size of battery required, or lengthen the device's running time. The design fits tiny magnets and wire coils onto keys. Each keystroke generates a small current that gets stored in a capacitor. The battery is slowly recharged from there. According to the inventor, this could potentially extend the battery life of a laptop to 10 hours—albeit after a lot of typing. The magnets do add weight, but this weight is small enough to be offset by lighter displays and other components.

The prospects of certain future technologies reaching the market depend heavily on power storage technologies for them being co-developed. A flexible screen, based on electronic ink, will require a power source comparable to itself. An Israeli company called Power Paper has developed a battery that is paper-thin. The anode and cathode are literally inks that are printed onto paper or other flexible materials. Unlike conventional cells, the power paper design requires no metal casing. The cell will provide 1.5 volts and is 0.5mm thick. Smart cards, disposable cell phones, intelligent clothing and even medical devices are possible applications of the technology.

Making devices faster and increasing functionality is what manufacturers stress on. But some are taking notice of the limitations of current battery technology, and are trying to counteract with devices that are smarter. Intel's Centrino, for example, is optimised for limited power consumption. Optimisation of power usage is the simplest way to increase battery time. As batteries continue to advance and workable fuel cells are made, your portable devices will become truly portable, freeing us from the worry of that dreaded battery dying. ■

KAIZAD VAJIFDAR

kaizad_vajifdar@thinkdigit.com

1/4 AD